

# VENNA RAJESH

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*Software Engineer with experience building scalable backend services, distributed systems, and full-stack applications using **Java**, **Go**, **Python**, **TypeScript**, and **React**. Experienced designing **REST APIs**, developing cloud-native applications, and delivering production-ready software through the full software development lifecycle using **Agile** and **CI/CD** practices.*

## Education

### California State University, Northridge

Master of Science in Software Engineering | GPA: 4.0/4.0

Dec 2025

Northridge, CA

## Experience

### California State University Northridge

Research Engineer

Northridge, CA

Aug 2024 – Present

- Developed telemetry pipelines over **HTTP/TLS/TCP/IP**, improving distributed system visibility from **4.2% to 10.5%**.
- Designed routing workflows to identify latency bottlenecks and improve network efficiency across distributed services.
- Implemented MILP optimization across a 42-node network, achieving **66.7%** flow acceptance under capacity constraints.

### Walmart (via Cognizant)

Software Engineer I

Mar 2022 – Aug 2023

- Owned production web workflows using **React**, **TypeScript**, and **Node.js**, delivering **8+** enterprise features from product requirements to release.
- Engineered **15+** reusable **React** components for forms, tables, modals, and shared UI flows, reducing duplicated frontend logic by **20–25%**.
- Integrated **10+** **REST API** workflows with typed frontend modules, standardizing validation, loading states, error handling, and service contracts.
- Reduced API response times by **10–15%** through **PostgreSQL** query tuning, payload cleanup, and request-path optimization.
- Implemented **PingFederate SSO** and RBAC permissions across **2** user access levels to secure authentication.
- Improved release reliability with **Jest** unit tests, peer code reviews, CI/CD validation, and defect triage across **6+** sprint releases.
- Debugged production issues across UI, API, and database layers, resolving **3–5** defects per sprint with QA, product, and backend.
- Tech Stack:** React.js, TypeScript, JavaScript, HTML, CSS/SCSS, Node.js, Express.js, PostgreSQL, REST APIs, Jest, Git, Docker, CI/CD, Jira, Confluence, Agile

### Walmart (via Cognizant)

Software Engineer Intern

Aug 2021 – Mar 2022

- Built full-stack features using **React**, **TypeScript**, **Node.js**, and **PostgreSQL** under senior engineer guidance.
- Developed responsive UI components from Figma designs and supported REST API integration for internal workflows.
- Contributed to unit testing, Git code reviews, CI/CD checks, and Agile sprint delivery.
- Tech Stack:** React.js, TypeScript, JavaScript, HTML, CSS, Node.js, Express.js, PostgreSQL, REST APIs, Jest, Git, Docker, CI/CD, Agile

## Projects

### Amazon Clone (E-Commerce System) | *React, TypeScript, Node.js, Express.js, PostgreSQL*

- Built a full-stack e-commerce system with **25+** reusable UI components for product discovery, cart, checkout, and orders.
- Designed **12+** **REST APIs** for auth, catalog, cart, and orders with structured validation and service-layer separation.
- Modeled **8** normalized **PostgreSQL** tables, reducing duplicate data across users, products, carts, and order workflows.

### Eye – AI Powered Diagnosis App | *PyTorch, ResNet, LangChain, Whisper, Gemini Pro*

- Trained a **ResNet** classifier with **92%** validation accuracy for retinal disease detection.
- Built a multimodal diagnosis pipeline combining image inference, voice input, and LLM reasoning across **3** AI services.
- Integrated **Whisper**, **Gemini Pro**, and **LangChain** to support real-time diagnosis conversations under **5s** response latency.

### E-Commerce Store and Admin Dashboard | *React.js, Node.js, Express.js, MongoDB, Redis, Stripe, Tailwind CSS*

- Built a production-style MERN platform with **20+** admin and customer workflows across catalog, cart, coupons, and checkout.
- Implemented **JWT** access/refresh tokens, protected routes, and role-based admin flows across **2** user permission levels.
- Integrated **Stripe** payments and **Redis** caching, reducing repeated product reads by **35%** in cached catalog flows.

## Technical Skills

|                    |                                                                                |
|--------------------|--------------------------------------------------------------------------------|
| Languages:         | Java, Python, C++, Go, TypeScript, JavaScript, SQL, Bash                       |
| Frontend:          | React.js, Angular, Next.js, HTML5, CSS3, Tailwind CSS, SCSS, Responsive UI     |
| Backend & APIs:    | Node.js, Express.js, REST APIs, Microservices, gRPC, JWT, OAuth, RBAC          |
| Databases:         | PostgreSQL, MongoDB, Redis, Query Optimization, Schema Design                  |
| Cloud & DevOps:    | AWS, Docker, Git, GitHub Actions, CI/CD, Linux, Deployment Pipelines           |
| Testing & Quality: | Jest, Unit Testing, Integration Testing, Code Reviews, Debugging               |
| AI/ML:             | PyTorch, ResNet, Whisper, LangChain, Gemini Pro, Multimodal AI                 |
| Concepts:          | System Design, OOP, Data Structures & Algorithms, Design Patterns, Agile, SDLC |

## Leadership & Activities

Mentoring: Practicum Mentor (CSUN), Code the Dream Mentor    Ambassador: Adobe Student Ambassador, Perplexity Student Ambassador    Hackathons: TreeHacks Participant    Awards : Star of the Month (3 critical sprint releases), AWS Certified Cloud Practitioner, Microsoft Azure AZ-900, Machine Learning & Deep Learning Specialization (Coursera)